

Siman Shrestha

Senior Software Engineer

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Senior Backend Engineer with 5+ years of experience building scalable service platforms and APIs. Experienced in distributed systems, microservices architecture, and observability. Relocating to NYC by December 2025.

TECHNICAL COMPETENCIES

Programming Languages: Java, C++, Python, TypeScript, JavaScript, Go

Frameworks: Spring Boot, NestJS, Node.js, LangChain, LangGraph, React, Angular

Databases: MySQL, Postgres, MongoDB, Redis, Hazelcast

Cloud & Tools: AWS, Docker, Kubernetes, Kafka, GraphQL, NGINX, CI/CD, GitHub Actions, MCP, Grafana

EXPERIENCE

Visa | Austin, TX - Hybrid

Senior Backend Software Engineer

Sep 2024 – Present

- Core backend developer for the Visa Accept product announced at Product Drop 2025, which enables micro merchants to accept payments on their mobile devices. Built scalable payment APIs to handle 1M+ merchants and process \$1B annually by FY26 across 50 countries, using Java/Spring Boot/MySQL/Hazelcast.
- Designed network affinity routing policies to eliminate cross-region database calls and ensure data consistency.
- Implemented blue-green deployment failover, achieving zero-downtime releases and 99.99% uptime.
- Secured APIs with HMAC signing, MTLS, and MLE for authentication and payment confidentiality.
- Developing Visa Accept Agent using GPT-4o/Python/LangChain to transform payment data into business intelligence for micro merchants, achieving SVP endorsement.
- Enhanced debugging efficiency by 85% (45 minutes to 5 minutes) by building an AI troubleshooting system using Python/LangChain/ChromaDB/GPT-4o to search through documentation, application logs, and database entries.

Hinge Health | Austin, TX - Remote

Backend Software Engineer II

Aug 2023 – Jul 2024

- Key contributor to microservices that manage sensitive health data for 30M patients and serve as the source of truth for all user data at Hinge Health using TypeScript/NestJS/Postgres.
- Accelerated patient outreach by 2.5 days and drove 6% user registration growth by redesigning the insurance data ingestion pipeline, achieving a 40% increase in throughput by leveraging AWS MSK (Kafka).
- Reduced service load by 30% by implementing an asynchronous job queue (BullMQ/AWS Elasticache/ Redis).

Visa | Austin, TX - Hybrid

Senior Fullstack Software Engineer - Observability

Jan 2022 – Aug 2023

Fullstack Software Engineer - Middleware

Oct 2020 – Jan 2022

- Built a real-time metrics dashboard serving 500+ reliability engineers, reducing server load by 5x and mean time to detection for production failures using React/Node.js/Redis/Go/GraphQL subscriptions via WebSockets.
- Owned an in-house observability platform used by 1000+ engineers to enable application monitoring, alerting, and logging using React/Node.js/MongoDB.
- Led deployment of Grafana Enterprise in a dual datacenter setup to maintain 99.99%+ uptime for critical payment APIs used by 1000+ engineers.

EDUCATION

Texas A&M University, College Station, TX

May 2020

Bachelor of Science in Computer Science, Minor in Visualization